Two ways a test population can change

Following the proposal feedback: broaden the mechanism, not just the list of corruptions.

01 IMAGE CORRUPTION

Change how an image looks.

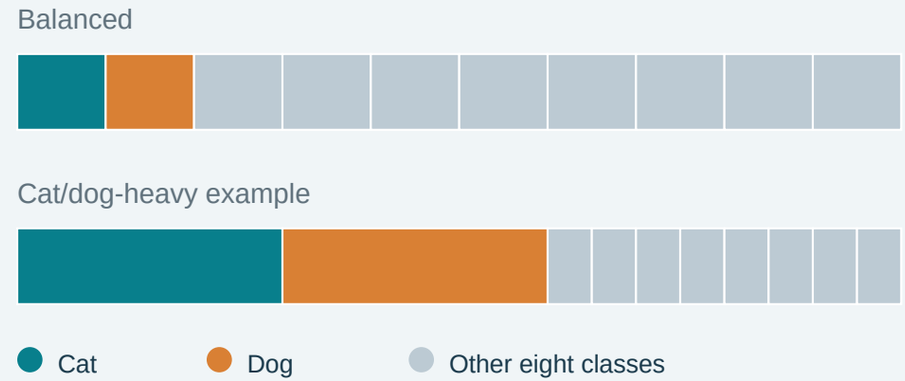
Clean Corrupted

Same semantic label; altered pixels.

Noise, blur, brightness.
The class mix is held fixed; prediction sets may change because the inputs change.

02 LABEL SHIFT

Change which classes appear more.



Same within-class images and per-image sets.
Only their contribution to the population changes.

DESIGN QUESTION

Does adding class-composition shift provide a useful contrast to image corruption?

Illustrations show the design, not measured results. Pixel tiles are schematic, not CIFAR-10 examples.

Label-preserving corruption does not by itself establish the formal assumption that P(Y | X) is unchanged.

How label-shift evaluation works

Calibrate once on clean data. Keep the predictor fixed. Reweight the evaluation summaries.

CALIBRATION BRANCH

5,000 clean calibration images → Choose threshold for 90% target → **Freeze model + threshold**

EVALUATION BRANCH / A SEPARATE SET OF 5,000 CLEAN IMAGES

1 Build prediction sets

For each evaluation image, record its fixed set C(x).

True cat, set {cat, dog}: covered

2 Summarize each class

Estimate coverage and set size within each true class.

c(cat) = covered cats / all cats

3 Change class weights

Combine class summaries using a chosen target prior.

Coverage = sum over y of q(y) c(y)

WHAT IS FIXED?

Images within each class, model outputs, prediction sets, and calibration thresholds.

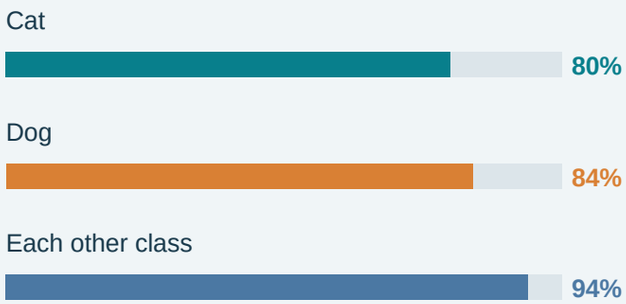
WHAT THIS EVALUATES

Sensitivity to class composition. Reweighting these metrics does not repair the prediction sets.

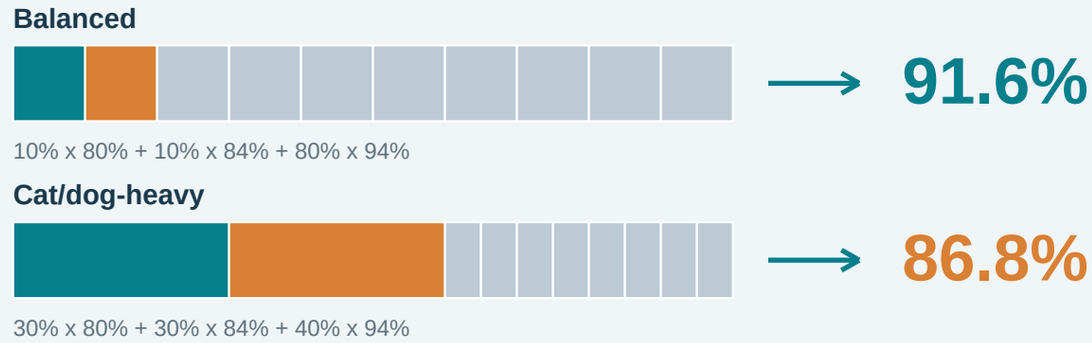
Why the class mix matters

ILLUSTRATIVE NUMBERS ONLY / This page contains no fitted-model results.

Assume fixed class coverages



Only change the population weights



The sets did not get worse. Classes with lower coverage became more common.

The direction can reverse if the emphasized classes have higher coverage.

DISCUSSION CHOICE: WHICH CLASSES SHOULD BECOME MORE COMMON?

Two interpretable examples

Cat + dog; airplane + automobile.
Easy to explain, but limited in scope.

All 45 class pairs

Same shift strength, different emphasized classes.
Show the range of outcomes, not only the mean.

FEEDBACK REQUESTED

Are two examples enough for this project, or should the analysis test sensitivity across more class pairs?